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An Intelligent Clinical Decision Support System for Heart Disease Diagnosis using Machine Learning

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Abstract. Heart disease prevails as the foremost cause of fatality, imposing a significant strain on individuals, families, and healthcare infrastructures globally. With the swift progress in technology, machine learning presents a promising avenue to tackle the challenges by forecasting the risk of heart disease before it becomes critical. The intention is to leverage the capabilities of ML algorithms to analyze patient data and uncover patterns associated with heart ailments. The proposed CDSS integrates the predictions of two data models based on clinical data and ECG data into a decision fusion module that highlights the key features contributing to forecasts in each model. This information is critical for interpretability and helps healthcare professionals understand the rationale behind the decision, which is the final result's dominating feature. It reduces the chances of misdiagnosis by combining multiple data sources and providing a clear risk assessment that aids in prioritizing care.

Keywords: Heart Disease, Electrocardiogram (ECG), Clinical Decision Support System (CDSS), Fusion Module, Risk Score.

1 Introduction

Cardiovascular Disease (CVD), often referred to as heart disease, represents a significant global health issue, comprising various conditions that impact the blood vessels of the heart. The basic categories include ischemic heart diseases and strokes, both of which present serious risks to human life. The World Health Organization (WHO) highlights several key behavioural risk factors that contribute to CVD, including poor dietary choices, lack of physical activity, tobacco consumption, and excessive alcohol intake. Prolonged exposure to these risk factors can lead to early warning signs like increased blood glucose and lipid levels, high blood pressure, obesity

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etc. Additional symptoms, as noted by the American Heart Association, may consist of chronic fatigue, difficulty breathing, swelling in the ankles or feet, and cognitive impairments. To effectively address the rising incidence of CVD, timely diagnosis and intervention are essential. Early identification can significantly reduce risks by allowing healthcare professionals to implement suitable treatments and lifestyle modifications. Machine learning has appeared as a valuable asset in the medical domain, capable of analysing intricate datasets and predicting outcomes with impressive precision. In this initiative, a machine learning-driven strategy has been used to assess the probability of CVD in individuals by utilizing essential health data and ECG information through deep learning techniques. The aim of this CDSS utilizes data-driven methodologies to evaluate an individual's likelihood of developing heart disease. By examining various elements such as medical history, lifestyle choices, demographic information, and clinical test outcomes, the system seeks to deliver precise risk assessments that facilitate early identification and prevention of heart disease. The proposed fusion model integrates multiple approaches to provide an efficient and accurate decision support system for healthcare professionals that offers a comprehensive analysis, combining the strengths of both predictive models to enhance diagnostic accuracy. The machine learning model trained on clinical data focuses on parameters such as cholesterol levels, chest pain type, and maximum heart rate, while the deep learning model processes ECG graphs to detect abnormalities like irregular heart rates. These two models are fused through a decision fusion module, which calculates a unified risk score and classifies patients into low, medium, or high-risk categories to ensure robust predictions and excellent reliability. This CDSS can discover the rationales behind the results produced by the system. It can also justify the reason behind its decision which are the main contributing factors for the outcome.

2 Literature Survey

Lassoued H. et al. [1] explore feedforward neural networks for ECG decision support, achieving high accuracy rates and optimal network structures for identifying cardiac arrhythmias, setting a benchmark for future research. B. Panganiban E. B. et al. in [2], proposes a CNN framework for ECG arrhythmia detection and classification, achieving high accuracy rates and enabling remote diagnosis through low-powered wearable biosensors. In [3], the authors present an ECG-based DSS for managing clinical data using ML algorithms to predict heart disease, enhancing healthcare professionals' accuracy and reliability. Pooja R. et al. in [4] designs a hybrid ML system for early heart disease prediction, utilizing multivariate imputation, data preprocessing and feature selection techniques for enhanced reliability. Adewole K. S. et al. in [5], review the development and application of DSS and expert systems in ECG diagnosis and explanation, highlighting advancements, challenges, and open research issues. Simegn G. L. et al. [6] highlight the significance of AI and machine learning in diagnosing cardiovascular diseases, proposing a system comprising an ECG processor, prediction module, and classification module. In [7], the authors classify Ventricular Arrhythmias with ECG images using machine learning algorithms successfully. Javeed A. et al. in [8], discuss machine learning techniques for predicting cardiac patient mortality,

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highlighting global heart disease prevalence, diagnostic systems, feature selection, class imbalance solutions, and performance metrics. Sherly S. I. et al. in [9], propose an ECG Based Multimodal Framework for heart disease detection with clinical data used to select the weight parameters in the HDCNN. In [10], the authors suggest that CNNs can be put to good use in classifying ECG signals and predicting CVD in a precise manner. Sankar P.N. et al. in [11], establish that Autoencoder, CNN, and LSTM methods produce good results in the accurate prediction of heart disease. In [12], Golande A. L. et al. propose a hybrid ECG-based prediction system in which FSN with LSTM performs well. Bozyel S et al. [13], emphasize the significance of AI in clinical decision support systems for cardiovascular disease, highlighting its potential to improve diagnosis, treatment, and management. Ahmed M. A. et al. in [14], presents an extensive overview of ML and DL techniques applied in DSS for heart disease prognosis.

2.1 Research Gaps

Existing Decision Support System (DSS) focusing on heart disease prediction incorporates ECG data or is implemented using clinical data. So, there is an exigency for the integration of multiple data inputs to collect, analyze, and interpret data, thereby assisting healthcare professionals in making decisions. An explainable recommendation system is much needed to provide actionable insights and enhance the understandability of the rationale behind the decision.

2.2 Novelties

The proposed fusion CDSS model improves the accuracy and consistency of decisions by analyzing multimodal types of data quickly and effectively. It combines the predictions of two approaches based on Clinical and ECG data to provide a unified result that offers risk scores and recommendations to assist doctors in diagnosing heart disease. It improves decision accuracy by reducing the chances of misdiagnosis by combining multiple data sources and providing a clear risk assessment. This system also helps to categorize patients as low, medium, or high risk, which aids in prioritizing care with explainable predictions.

3 Methodology

In this work, the proposed CDSS is trained with various ML models like LR, SVM, Decision Tree, KNN, Naïve Bayes, Random Forest etc. that combine clinical and ECG data models into a Decision Fusion model depending upon risk factors for achieving more effective and precise results.

The following block diagram depicts the working process of the CDSS.

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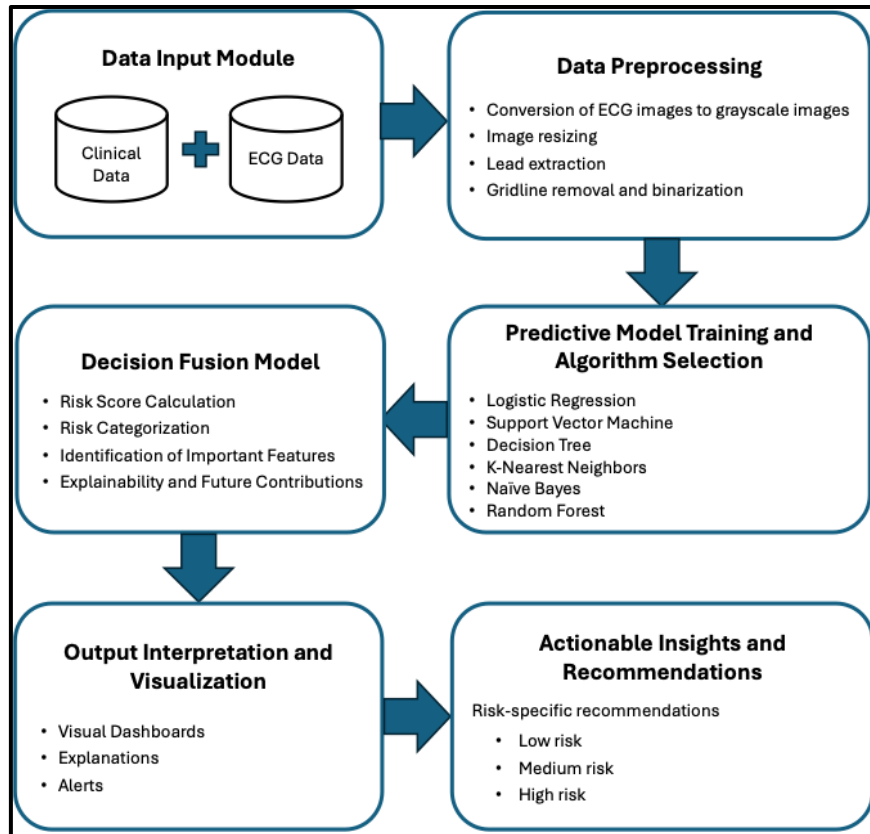


Fig. 1. The working process of the CDSS

The working process of the proposed CDSS is divided into six sections.

- A. Data Input Module
- B. Data Preprocessing
- C. Predictive Model Training and Algorithm Selection
- D. Decision Fusion Model
- E. Output Interpretation and Visualization
- F. Actionable Insights and Recommendations

A. Data Input Module

The data input module serves as the foundation for the entire CDSS by ensuring high-quality and usable data. It comprises two sub-modules:

- **Clinical Data module:** Includes patient attributes such as cholesterol levels, maximum heart rate, blood pressure, age, and other clinical parameters. The

dataset used here collected from Kaggle¹, contains 1000 different patient's clinical data for 13 different parameters.

- **ECG Data Module:** The dataset used in this work is sourced from the Mendeley² Data Repository. It includes graphs or extracted features from ECG signals categorized into four distinct classes:
 - a) *Normal*: ECG signals representing a healthy heart.
 - b) *Myocardial Infarction*: Images indicating a heart attack.
 - c) *Abnormal Heartbeat*: Signals representing irregular heart rhythms.
 - d) *History of Myocardial Infarction*: Previous occurrences of heart attacks captured in ECG data.

The dataset contains 240 ECG Images of Patients with Myocardial Infarction, 233 ECG Images of Patients with Abnormal Heartbeats, 172 ECG Images of Patients who have History of MI and 284 Normal Person ECG Images.

B. Data Preprocessing

This module ensures that data from both sources are consistently formatted and ready for analysis by the predictive models. It is critical for converting raw ECG images into a format suitable for machine learning. The following steps are applied:

- a) **Conversion to Grayscale:** ECG images are transformed into a grayscale format to simplify image data by reducing it to intensity levels.
- b) **Image Resizing:** Images are resized to standardized dimensions required for the analysis.
- c) **Lead Extraction:** Each image is divided into 12 sections to extract individual leads (1-12). These leads correspond to the 12-lead ECG system, essential for understanding heart functionality.
- d) **Gridline Removal and Binarization:** Gridlines are removed to isolate the signal. Images are then converted into binary format to enhance contrast and focus solely on the signal.

C. Predictive Model Training and Algorithm Selection

The work involves testing multiple ML algorithms to recognize the best-performing model for ECG-based cardiovascular disease classification. The ensemble approach is used to improve accuracy by combining the strengths of different algorithms. The predictive models are the core analytical engines of the CC. They analyse the input data and generate predictions.

- **Predictive Model based on Clinical Data:** A machine learning model is trained on clinical parameters.
- **Predictive Model based on ECG Data:** A deep learning model is trained on ECG graphs.

Each model focuses on analysing its respective data source to provide robust predictions.

¹<https://www.kaggle.com/datasets/jocelyndumlao/cardiovascular-disease-dataset>

²<https://data.mendeley.com/datasets/gwbz3fsgp8/2>

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D. Decision Fusion Model

The decision fusion module combines the outputs of the clinical and ECG models to generate a unified decision. This module ensures that the strengths of both models are utilized for more accurate predictions. Key steps in decision fusion include:

- a) **Risk Score Calculation:** The probabilities from both models are combined using a weighted average:

$$\text{Risk Score} = w_1 \cdot P_{\text{clinical}} + w_2 \cdot P_{\text{ECG}}$$

- 1) w_1, w_2 : Assigned weights based on model accuracy or reliability as the more accurate and reliable model contributes more to the final decision.
 - 2) $P_{\text{clinical}}, P_{\text{ECG}}$: Probability scores are achieved from the clinical and ECG models that define the likelihood that the input belongs to a particular class.
- b) **Risk Categorization:** Based on the calculated risk score, patients are classified into three risk categories:
 - 1) *Low Risk*: Risk Score < 0.3
 - 2) *Medium Risk*: $0.3 \leq \text{Risk Score} < 0.7$
 - 3) *High Risk*: Risk Score ≥ 0.7
 - c) **Identification of Key Features:** It can be recognized by the Gini index of the Random Forest model as illustrated in Fig. 2.
 - 1) *ECG Model*: Abnormal heart rate is a key feature extracted from the ECG signal that strongly influences the CVD prediction
 - 2) *Clinical Model*: Slope and chest pain are identified as the most critical clinical parameters for determining heart disease risk.
 - d) **Explainability and Feature Contributions:** The CDSS provides insights into how much each feature contributed to the overall risk score.

E. Output Interpretation and Visualization

This component translates the predictions and risk scores into user-friendly outputs for healthcare professionals. It includes:

- a) **Visual Dashboards:** Displays key information such as the risk score, risk category, and contributing factors in a clear and concise manner.
- b) **Explanations:** Provides interpretability for the predictions by highlighting which clinical parameters or ECG features contributed most to the decision.
- c) **Alerts:** Generates alerts for high-risk patients, ensuring timely medical attention.

F. Actionable Insights and Recommendations

The CDSS provides actionable recommendations to guide healthcare professionals in patient management. These include:

- a) **Risk-Specific Recommendations:** Low Risk: Suggest lifestyle changes and routine monitoring.
- b) **Medium Risk:** Recommend further diagnostic tests (e.g., stress tests, echocardiograms).
- c) **High Risk:** Advise immediate medical intervention or hospitalization.

4 Design of Decision Support System

This module involves integrating multiple components to collect, analyze and interpret data, thereby assisting healthcare professionals in making decisions from this twofold system. This work already has two predictive models: one using clinical data and another using ECG data. A CDSS will combine these models, provide actionable insights and enhance the decision-making process.

The whole system combines several steps from start to end including

- Data is collected for two models, e.g. clinical data and ECG report.
- The data is analyzed with the help of the ML model, trained by clinical data and the DL model, trained with ECG graphs.
- Identification of the key features of the data.
- Then, the decision fusion model of CDSS generates a unified decision for more accurate predictions.
- Based on the decision fusion calculations, provide suggestions to the user.

Identifying the key feature quantifies how much it contributes to reducing the impurity in the decision-making process. Using the Gini index of the random forest model, the most significant feature identified in the clinical data is the slope, which is the slope of the ST segment; by analyzing this, doctors can potentially identify areas of coronary artery disease.

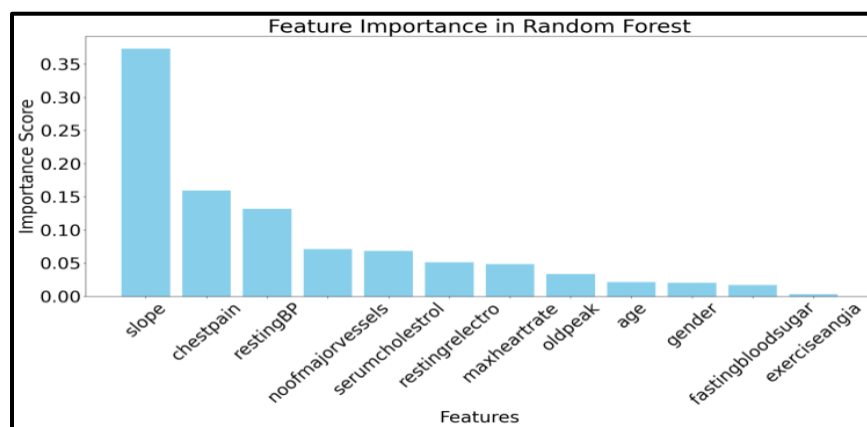


Fig. 2. Feature importance in Random Forest

The decision fusion model combines the outputs of the clinical and ECG models and collects the possibility of the person with heart disease, generating a risk score based on the accuracy of the model, which determines whether the person has heart disease or not. The probabilities from both models are combined using a weighted average: $\text{Risk Score} = w_1 \cdot P_{\text{clinical}} + w_2 \cdot P_{\text{ECG}}$. Based on the calculated risk score, patients are classified into three risk categories, as mentioned in the decision fusion model.

The final output from the CDSS consists of four types of incidents:

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- a) When both the models predict the patient having heart disease, and the risk score falls within the High-Risk Category depicted in Fig. 3, the CDSS strongly suggests the patient take immediate medical attention.

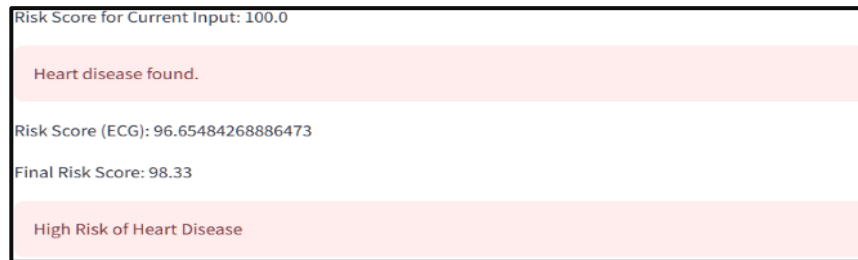


Fig. 3. High risk for both results

- b) When both the models predict the patient does not have heart disease, and the risk score falls within the Low-Risk Category as defined in Fig. 4, the CDSS suggests that the patient is not at risk of heart disease, but regular monitoring is still advisable.



Fig. 4. Low risk for both results

- c) When the ECG result predicts that the person does not have heart disease, but the clinical data predicts that the person has heart disease, the risk score is more likely to fall into the category of Mid-risk and the patient is advised to consult to doctor and recheck the reports as explained in Fig. 5.



Fig. 5. High risk for clinical results and Low risk for ECG

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- a) When the clinical data predicts that the person does not have heart disease, but the ECG result predicts that the person has heart disease, the risk score is more likely to fall into the category of Mid-risk and the patient is advised to consult to doctor and recheck the reports like Fig. 6.

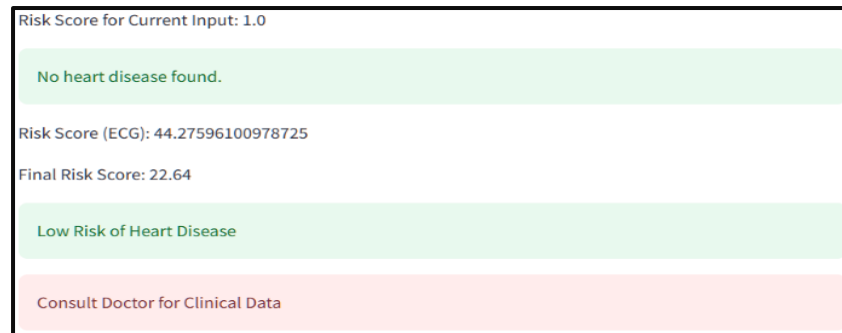


Fig. 6. Low risk for clinical results and High risk for ECG

5 Results And Discussion

In the preprocessing of an ECG image, first, it was converted into a Grayscale image shown in Fig. 7.



Fig. 7. Conversion of ECG image into Grayscale image

Then the Grayscale transformed image is divided into 12- lead sections. After that, the processed image is removed with gridlines and converted to binary. This process is depicted in Fig. 8.

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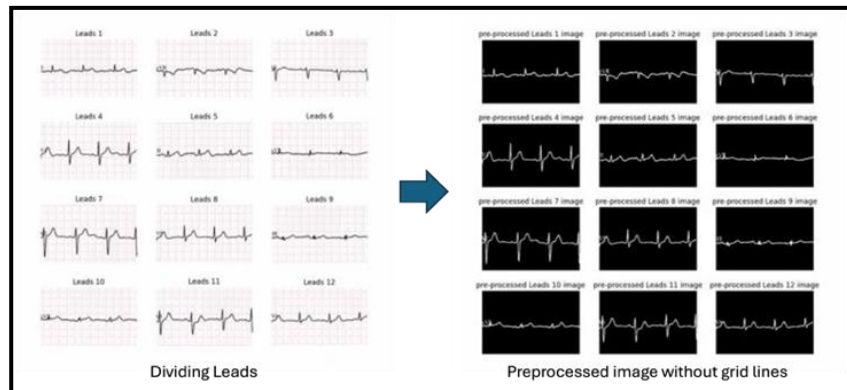


Fig. 8. Dividing the image into 12 leads and pre-processed the image by removing the gridlines and converted to binary

Next, contours are extracted from each ECG image to isolate the signal patterns from the background. After extracting the contours, the signal is converted into a one-dimensional (1D) array. The 1D signal values are normalized using MinMaxScaler, bringing all features into a consistent range. Scaling prevents even a single feature from controlling the ML model due to differences in magnitude. Signals from all 12 leads of the ECG are combined into a single structured dataset. A .csv file is created where each row corresponds to an ECG image, and the columns represent the scaled values of the 12 leads along with the target label (e.g. Normal, Abnormal). The whole process is shown in Fig. 9.

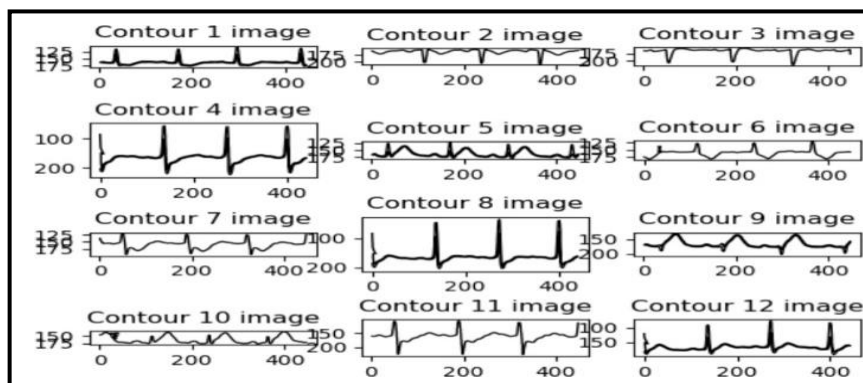


Fig. 9. Extraction of Contours for each lead (1-12)

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After converting the signals into a 1D array, 3059 features are received and then MinMaxScaler is performed to normalize. This step is depicted in Fig. 10.



Fig. 10. Dimensionality reduction after converting to 1D signal

For the predictive model training, the ML algorithms including SVM, Logistic Regression, Decision Trees, KNN, Random Forest and Naïve Bayes are used to train and build the best-performing model for heart disease prediction with risk score. The evaluation metrics used for comparison are precision, accuracy, recall, and F1-score, which are calculated on the test dataset. The comparison of evaluation metrics is summarized in Table 1.

Table 1. Comparison of Evaluation Metrics

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	96.3	96.2	96.3	96.2
Support Vector Machine	79.6	79.5	79.6	79.5
Decision Tree	96.1	96.1	96.1	96.1
K-Nearest Neighbors (KNN)	79.5	79.4	79.5	79.4
Naïve Bayes	94.6	94.6	94.6	94.5
Random Forest	97.6	97.6	97.6	97.6

The Random Forest model achieves 97.6 accuracy from the hyperparameter tuning to select the best model. Random Forest achieves 99.5% accuracy with CuML, due to CuML's GPU acceleration, which enhances the computational efficiency and model's performance and also 98% for standard implementation. The confusion matrices of each technique have been depicted in Fig. (11-16).

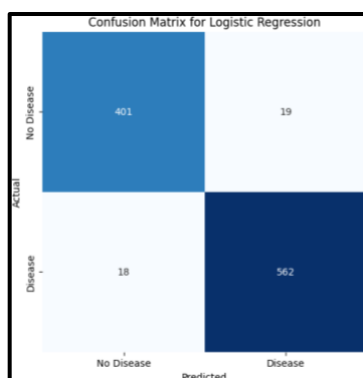


Fig. 11. Confusion Matrix for Logistic Regression

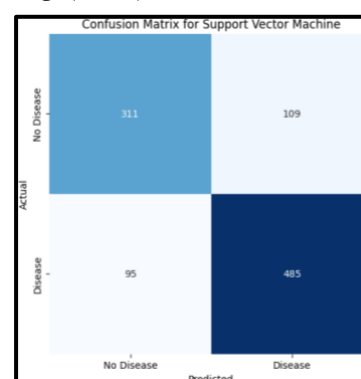


Fig. 12. Confusion Matrix for SVM

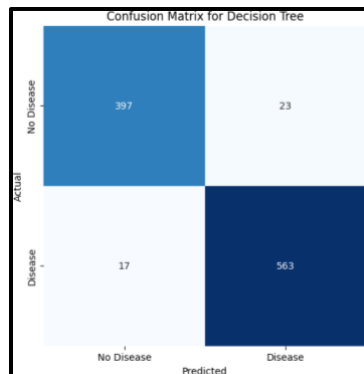


Fig. 13. Confusion Matrix for Decision Tree.

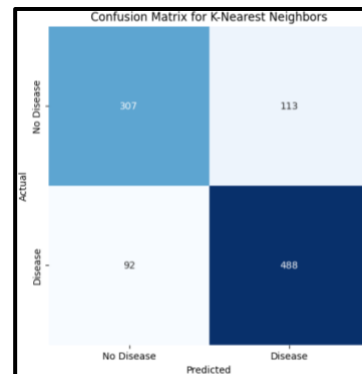


Fig. 14. Confusion Matrix for KNN

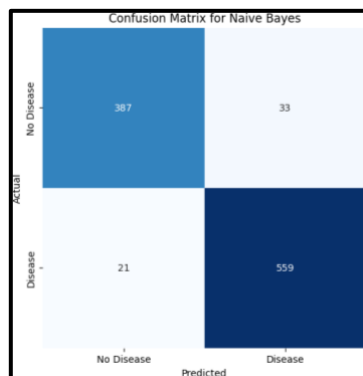


Fig. 15. Confusion Matrix for Naïve Bayes.

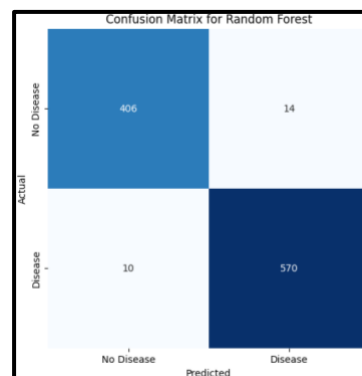


Fig. 16. Confusion Matrix for Random Forest

From Fig. (11-16), it can be clearly stated that the Random Forest gives the best result for the CDSS. With the implementation of CuML for Random Forest, it achieves 99.5% accuracy.

6 Conclusion

This Clinical Decision Support System (CDSS) not only provides a prediction of whether a patient has heart disease but also highlights the key contributing factors, enabling interpretability, explainability and transparency. The risk stratification mechanism ensures the timely identification of high-risk patients, allowing healthcare professionals to prioritize and take immediate action when necessary. Furthermore, actionable insights and recommendations generated by the CDSS assist in improving patient care and outcomes. This work demonstrates how the fusion of ML and DL can revolutionize healthcare diagnostics. An accurate, interpretable, and practical CDSS will significantly reduce the burden of heart disease by enabling

early detection and effective intervention. Future enhancements, such as incorporating additional data sources and refining the models, can improve its performance and applicability in real- world healthcare scenarios.

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